

YONATHAN WONDOSSEN

SENIOR SOFTWARE ENGINEER

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SUMMARY

Hey! I'm Yonathan Wondosen, a **Senior Software Engineer & Architect** specializing in enterprise-grade system design, Complex Web-applications Design and Development and microservices architecture. I architect and build scalable solutions using **React/Next.js, TypeScript, Python (FastAPI, Flask), Laravel, Node/Express**, and **SQL / NoSQL databases**.

I leverage **AI-assisted development** (Claude AI Sonnet 4.5/4.6) to accelerate delivery while maintaining code quality. My expertise spans **system design, scalable architectures, Docker/Kubernetes** and **PM2-managed deployments, VPS infrastructure**, and **CI/CD automation**. I excel at delivering complex solutions under high-pressure deadlines for international clients. Familiar with microservices patterns and distributed systems.

EDUCATION

BSc in **Computer Science**
UNITY UNIVERSITY
2022 - 2026
Addis Ababa, Ethiopia

SKILLS

- Project Management
- Teamwork
- Time Management
- Effective Communication

LANGUAGES

- English
- Amharic
- Oromo (basic)

WORK EXPERIENCE

■ FullStack Software Engineer

VELKAP Ltd - London, UK
2026 - Present . Contractual . Remote

- Developing enterprise investment management system for UK-based fintech company.
- Building full-stack investment management platform using Laravel and Next.js
- Architecting scalable backend APIs for Investment management and financial transactions
- Implementing responsive frontend with modern React patterns and TypeScript
- Collaborating with teams
- Ensuring compliance with UK financial regulations and data security standards

Problem: Implemented a co-investor feature allowing multiple investors to participate in a single investment, requiring complex ownership and investment allocation calculations.

Solution: Designed the underlying data models and business logic to manage participant relationships, automate investment share calculations, and ensure accurate financial tracking and reporting.

■ **AI Focus Back-end Engineer, RAG Development**

Akilo Technology - HongKong China

2025 - 2025 . Contractual . Remote

- AI Focus Back-end Engineer, RAG Development
- Worked on a RAG AI Chat bot Development that integrates a Large Language Model (LLM) with MongoDB vector search
- Used embeddings to store and retrieve relevant documents based on user queries with HNSW Algorithm
- The retrieved context is then passed to the LLM to generate accurate, context-aware responses

Problem: Identified and resolved low retrieval accuracy caused by suboptimal embeddings and vector search configuration.

Solution: Tuned HNSW parameters, improved embedding quality, and implemented efficient document chunking strategies. Enhanced semantic search relevance and reduced LLM hallucination, resulting in more accurate, context-aware responses.

■ **Fullstack Software Engineer**

EtPay-Rent PLC, Addis Ababa Ethiopia

2025 March - 2025 Nov . Full-time . Hybrid

- Led development and maintenance of a scalable web application, ensuring performance, reliability, and system stability.
- Developed and maintained a microservices-based system.
- Integrated Telebirr and multiple payment gateways, enabling secure and seamless digital transactions
- Diagnosed and resolved payment processing and system integration issues, improving transaction success rate and reducing failures
- Conducted technical research and solution evaluation to support system enhancements and integration decisions.

Problem: System performance degradation and bugs during ongoing development and maintenance

Solution: Refactored critical modules, optimized queries, and fixed bottlenecks to improve reliability and response time.

■ **Technology Officer**

Ethiopian Kidney Care NGO - Addis Ababa Ethiopia

2024 - PRESENT . **Volunteer**

- Led development of multiple high-impact digital projects and supported overall IT operations and system management.
- Designed and implemented a lightweight digital screening system, automating data collection and validation to improve accuracy, efficiency, and accessibility of campaign results.
- Implemented new features across existing web applications, improving usability and increasing user engagement
- Coordinated technical initiatives and contributed to system enhancements to improve organizational digital infrastructure

Problem: The NGO's digital systems and initiatives were fragmented, with limited technical structure, making it difficult to scale solutions, maintain consistency, and efficiently support operational and outreach activities.

Solution: Introduced structured development practices, unified system improvements across projects, and provided technical leadership to standardize workflows, improve reliability, and support scalable digital operations.

PROJECT 2025/26

- **AI Powered Electronic Medical Record System
BEGO Healthcare Solution**

Intelligent EMR platform designed for Ethiopian healthcare providers. Streamline patient care, manage multiple branches and unlock powerful insights.

Tech: React, Flask, REST API, Postgresql

- **RAG AI Chatbot**

Advanced RAG AI system integrating Large Language Models with MongoDB vector search for accurate, context-aware responses.

Tech: PythonFlask, OpenAI API , MongoDB, HNSW, Vector Search

- **EmployTrack - Enterprise HR System**

Full-stack enterprise HR management platform with multi-tenant architecture, role-based access control, attendance tracking, and leave management system.

Tech: Next.js, FastAPI, PostgreSQL, Docker, TypeScript

- **Property Managment ETpay Rent**

Modern property management solution with automated rent collection and maintenance tracking.

Tech: React, Java Spring Boot, PostgreSQL, REST API

- **Better auth - oAuth Implementation**

Implementing secure social login with better auth to fully protected OAuth login, automatic session handling, and seamless integration with my backend and Prisma Schema.

Tech: Node, Next, Prisma, oAuth, Better Auth

- **AI Driven MicroFinance Solution**

AI-driven credit scoring system using verified identity and behavioral data.

Tech: Python, Flask, React, AI, REST API, JWT,

- **Chapa Payment Gateway Integration**

Seamless Ethiopian payment gateway integration with comprehensive transaction management.

Tech: Laravel PHP, MySQL, CHAPA_API, REST API